

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Advanced Java Programming**

Code No: **CACS354** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Why do we use Panels while creating GUI applications in Java? Explain the components of the event handling model in Java.
3. What are adapter classes? What is the benefit of using adapter classes in Java while creating events in GUI programs? Explain with an example.
4. How the data can be accessed through a resultset object? Explain forward only, scroll-insensitive and scroll-sensitive resultsets.
5. What is a Javabean? Explain different types of properties in Javabeans.
6. What are the differences between servlets and Java applications? Explain different methods used for handling cookies in Java.
7. What are marshalling and unmarshalling of arguments in RMI? Differentiate between CORBA and RMI.
8. Write short notes on (Any Two): a) 2D shapes in swing b) MouseEvent and MouseListener c) Session tracking

Group C

Attempt any TWO questions[2x10=20]

9. Why do we need to use Layout managers while creating GUI application in Java? Explain the key interfaces required to develop any database application using Java. Also, write a program to display the records from the table Students who have got distinction and have a major subject Data science. Assume that the Student table is in the database College and contains the columns Rollno, Name, Level, Division and Major.
10. Differentiate between declaration and expression tags used in JSP. Also create a JSP page that contains three textboxes to input three numbers and a button. It should display the greatest number among three numbers when the button is clicked.
11. What is RMI? Explain its components. Write a client-server application in RMI to find the selling price of an item with cost price Rs. 5000 after a discount of Rs. 50.